AI REASONING SYSTEM

Project Report

1. CHOSEN REPRESENTATION

The proposed AI reasoning system uses symbolic knowledge representation to represent facts, rules, and queries in a structured format.

For the RuleTaker dataset, facts are represented using predicate-argument tuples. For example, the statement "Bob is young" can be represented as:

('young', ('Bob',))

Rules are represented using conditions and conclusions. Variables such as ?x are used to match different entities in the knowledge base.

For the FOLIO dataset, first-order logic (FOL) is used to represent knowledge using predicates, variables, quantifiers, conjunction, disjunction, implication, and negation.

For the CounterBench dataset, causal relationships are represented using a causal-graph-based approach to reason about counterfactual questions.

The symbolic representation allows the system to perform logical reasoning using explicit facts and rules.

2. REASONING ALGORITHMS USED

2.1 Forward Chaining

Forward chaining is used for reasoning on the RuleTaker dataset.

The system starts with the available facts and checks whether the conditions of a rule are satisfied. If all the conditions match, the conclusion is derived and added to the knowledge base.

This process continues until no new facts can be generated. Finally, the system checks whether the given query is supported by the derived facts.

2.2 First-Order Logic Resolution

First-order logic resolution is used for logical inference experiments on the FOLIO dataset.

The NLTK ResolutionProver is used to determine whether a conclusion logically follows from the given premises.

For example:

- Every human is mortal.
- Socrates is human.
- Therefore, Socrates is mortal.

The prover uses the logical premises to establish whether the conclusion can be derived.

2.3 Causal and Counterfactual Reasoning

A causal-graph-based approach is used as a preliminary baseline for the CounterBench dataset.

The system examines causal relationships between variables and evaluates questions involving interventions, such as whether an outcome would occur if a particular cause were changed.

3. EXPERIMENTS PERFORMED ON PUBLIC DATASETS

The system was evaluated using three public reasoning datasets: RuleTaker, CounterBench, and FOLIO.

The following table summarizes the recorded experiments.

Dataset	Number of Examples	Accuracy	Runtime
RuleTaker (Development Subset)	1000	89.7%	0.439 seconds
CounterBench Alpha V1	1000	40.0%	0.144 seconds
CounterBench Backdoor V2	200	52.5%	0.094 seconds
FOLIO Validation	203	86.7%	0.085 seconds

RuleTaker: The system achieved 89.7% accuracy on a subset of 1,000 development examples using forward chaining.

CounterBench: The preliminary baseline achieved 40.0% accuracy on Alpha V1 and 52.5% accuracy on Backdoor V2. These experiments examine causal and counterfactual reasoning.

FOLIO: The dataset contains three possible labels: True, False, and Uncertain. First-order logic representation and NLTK resolution were used in the reasoning experiments.

The recorded results demonstrate the application of symbolic reasoning techniques across different types of reasoning datasets.

4. EXAMPLES OF REASONING TRACES

Example 1: Forward Chaining

Initial Facts:

- Bob is young.
- Bob is kind.

Rules:

- If someone is young, they are smart.
- If someone is smart and kind, they are successful.

Reasoning Steps:

Step 1: The system checks the rule:

young(x) → smart(x)

Since Bob is young, the system derives:

smart(Bob)

Step 2: The system checks the second rule:

smart(x) AND kind(x) → successful(x)

Since Bob is both smart and kind, the system derives:

successful(Bob)

Final Query:

Is Bob successful?

Result: TRUE

The system derives the conclusion by applying the rules to the available facts.

Example 2: First-Order Logic Resolution

Premises:

1. Every human is mortal.
2. Socrates is human.

Query:

Is Socrates mortal?

Reasoning:

The universal rule states that every human is mortal. Since Socrates is a human, the rule can be applied to Socrates.

Therefore:

Mortal(Socrates)

Result: TRUE

The NLTK ResolutionProver successfully returned True for this test.

Example 3: CounterBench Causal Reasoning

Given causal relationships:

- Ziklo causes Blaf.
- Blaf causes Trune.
- Trune causes Vork.

- Vork causes Lumbo.

Question:

Would Lumbo occur if not Ziklo instead of Ziklo?

Dataset Answer: NO.

The preliminary baseline predicted NO for this example.

This illustrates how causal relationships are used to examine a counterfactual question.

5. CONCLUSION

The project explores the development of a symbolic AI reasoning system using structured knowledge representation and classical reasoning techniques.

Forward chaining is used to derive new facts from existing knowledge, while first-order logic resolution is explored for logical inference. A causal-graph-based baseline is also implemented for counterfactual reasoning.

The experiments demonstrate the application of these reasoning techniques to RuleTaker, CounterBench, and FOLIO datasets.

The results highlight the importance of symbolic reasoning, logical inference, and structured knowledge representation in developing general-purpose AI reasoning systems.